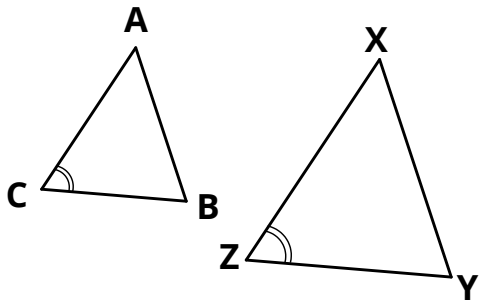
Proving triangles similar

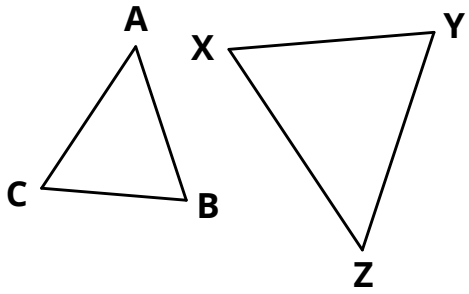
1 Use the information given in the diagram to complete the two-column proof that shows that $\triangle ABC \sim \triangle XYZ$.



To prove: $\triangle ABC \sim \triangle XYZ$

	Statements	Reasons
1.	$\frac{AC}{XZ} = \frac{BC}{YZ}$	Given
2.	$\angle ACB \cong \angle XZY$	Given
3.	$\triangle ABC \sim \triangle XYZ$	▯

2 Use the information given in the diagram to complete the two-column proof that shows that the two triangles are similar.

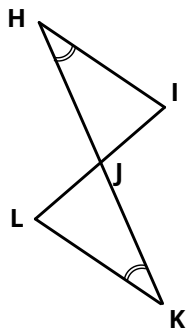


To prove: The two triangles are similar.

	Statements	Reasons
1.	$\frac{AC}{YZ} = \frac{AB}{XZ} = \frac{CB}{XY}$	Given
2.	▯	▯

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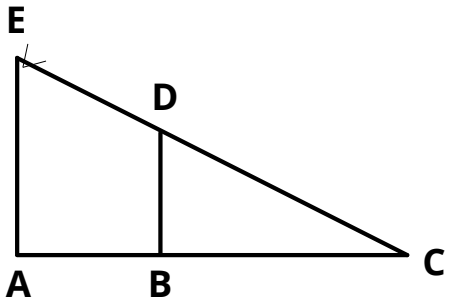
3 Use the information given in the diagram to complete the two-column proof that shows that the two triangles are similar.



To prove: The two triangles are similar.

	Statements	Reasons
1.	$\angle IHJ \cong \angle LKJ$	Given
2.	$\angle IJH \cong \angle LJK$	Vertical angles congruence theorem
3.	▯	▯

4 Use the information given in the diagram to complete the two-column proof that shows that the two triangles are similar.

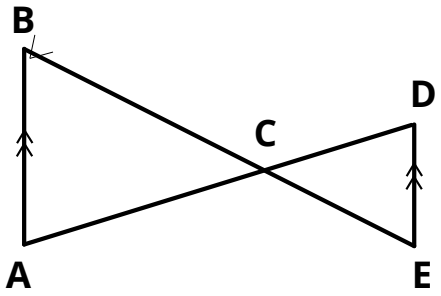


To prove: The two triangles are similar.

	Statements	Reasons
1.	$\frac{AC}{BC} = \frac{EC}{DC}$	Given
2.	$\angle ACE \cong \angle BCD$	Reflexive property of congruence
3.	▯	▯

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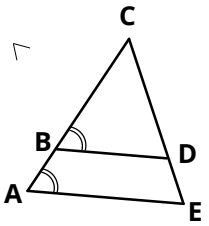
5 Use the information given in the diagram to complete the two-column proof that shows that the two triangles are similar.



To prove: The two triangles are similar.

	Statements	Reasons
1.	$\overline{AB} \parallel \overline{DE}$	Given
2.	$\angle BAC \cong \angle EDC$	Alternate interior angles theorem
3.	$\angle DCE \cong \angle ACB$	Vertical angles congruence theorem
4.	▯	▯

6 Use the information given in the diagram to complete the two-column proof that shows that $\triangle ACE \sim \triangle BCD$.

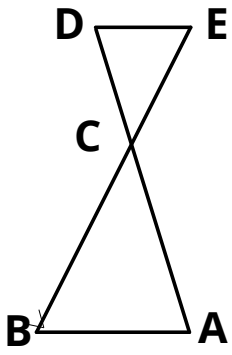


To prove: $\triangle ACE \sim \triangle BCD$

	Statements	Reasons
1.	$\angle CBD \cong \angle CAE$	Given
2.	▯	▯
3.	$\triangle ACE \sim \triangle BCD$	Angle-angle similarity (AA $\sim$)

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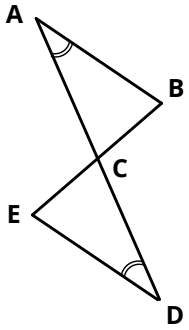
7 Use the information given in the diagram to complete the two-column proof that shows that $\triangle ABC \sim \triangle DEC$.



To prove: $\triangle ABC \sim \triangle DEC$

	Statements	Reasons
1.	$\frac{BC}{CE} = \frac{AC}{CD}$	Given
2.	▯	▯
3.	$\triangle ABC \sim \triangle DEC$	Side-angle-side similarity (SAS $\sim$)

8 Use the information given in the diagram to complete the two-column proof that shows that $\triangle ABC \sim \triangle DEC$.



To prove: $\triangle ABC \sim \triangle DEC$

	Statements	Reasons
1.	$\angle BAC \cong \angle EDC$	Given
2.	▯	▯
3.	$\triangle ABC \sim \triangle DEC$	Angle-angle similarity (AA $\sim$)

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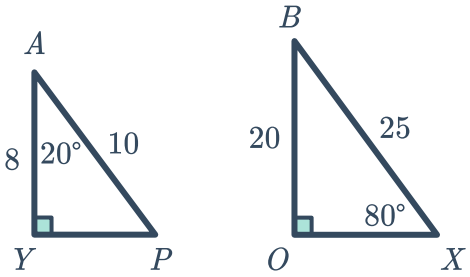
Additional questions

9 Determine whether each statement is true or false:

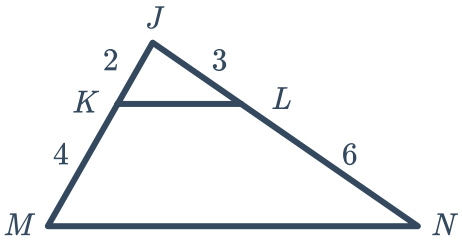
- **a** If two angles of one triangle are congruent to two angles of another triangle, then the triangles are similar.
- **b** If two sides of one triangle are proportional to two corresponding sides of another triangle, then the triangles are similar.
- **c** If two sides of one triangle are congruent to two corresponding sides of another triangle, then the triangles are similar.
- **d** If two triangles are similar, then their corresponding sides are congruent.
- **e** If two triangles are similar, then their corresponding angles are congruent.

10 Determine whether each pair of triangles is similar. If so, write a similarity statement and justify with a similarity postulate or theorem. If not, explain why not.

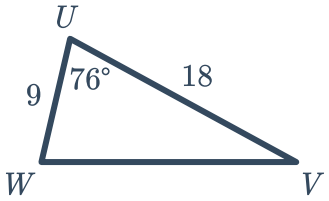
a

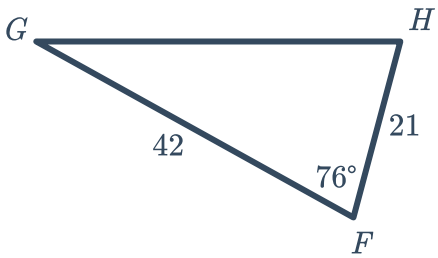


b

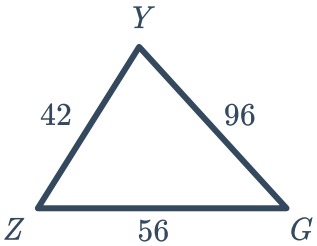


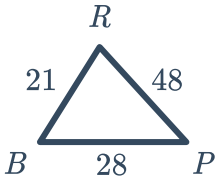
c





d

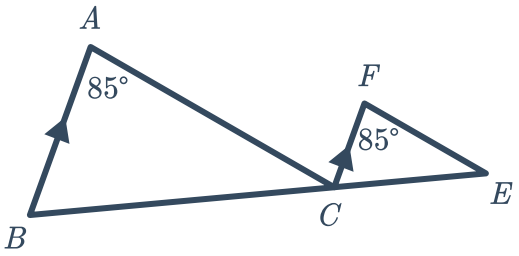




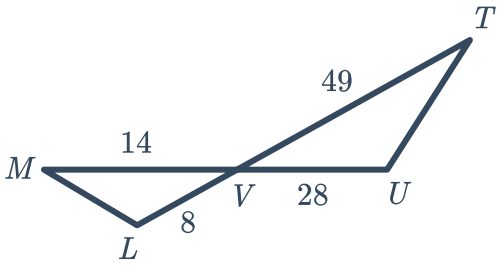
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11 Show that each of the following pairs of triangles are similar by identifying the similarity postulate or theorem and listing the pairs of sides and angles that justify it.

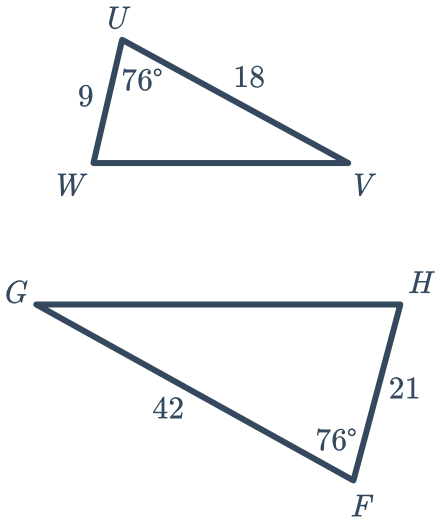
a



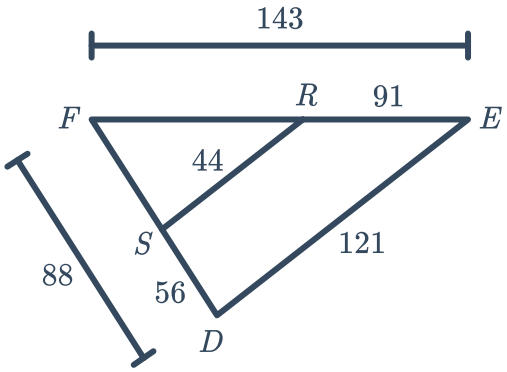
b



c

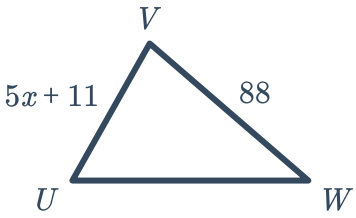


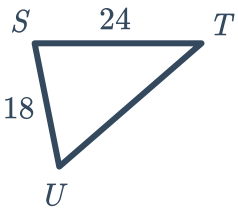
d



12 Given: $\angle S \cong \angle V$

Determine the value of x that makes $\triangle VUW \sim \triangle SUT$.





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